

Arnav Surve

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EDUCATION

California State University - Sacramento

Bachelor of Science, Computer Science

Sacramento, CA

Aug. 2024 – May 2026

Data Structures & Algorithms, Systems Programming, x86 Computer Architecture, Java OOP

Purdue University

Bachelor of Science, Cybersecurity – Dean's List, Fall 2023

West Lafayette, IN

Aug. 2022 – May 2024

Cybersecurity Fundamentals I & II, System Administration

SKILLS

Languages: Python, Go, PostgreSQL, MySQL, Javascript, HTML/CSS, Java

Frameworks: Echo, GORM, FastAPI, SQLAlchemy, Pydantic, Pytest, React, Node.js, Next.js, Agile Scrum

Tools: Git, GitHub, Vim, AWS, Docker, Redis, Nginx, Makefile, Prometheus, Grafana

Related: Debian/RHELinux, Software Engineering, DevOps, Distributed Systems, Cloud Computing, Version Control, Documentation, Testing, Back-End, Full-Stack

EXPERIENCE

IT Infrastructure Intern

June 2024 – Aug. 2024

MITER Brands – Milgard Manufacturing

Tacoma, WA

- Delivered a comprehensive proposal to optimize monitoring pipelines, reducing average alert redundancy by 60%
- Cross-communicated between 6 teams across 2 business units to gather insights and consensus on optimization initiative.
- Developed monitors for REST API endpoints, identifying discrepancies resulting in an 85% decrease in latencies.
- Analyzed monitoring groups, escalation policies, and alert thresholds to address tech debt and improve efficiency by 20%.
- Automated compliance audits for Active Directory controlled accounts using PowerShell scripting.

Undergraduate Research Assistant

March 2024 - May 2024

NASA California Space Grant Consortium

Sacramento, CA

- Constructed an infrasound frequency monitor using Raspberry Pi, enabling real-time aggregation of seismic and atmospheric activity with a 95% uptime in deployment.
- Implemented k-means clustering and autoencoder neural networks using scikit-learn and Tensorflow on over 25 million data points, predicting volcanic activity with a silhouette score of 0.47.

Software Engineer

Aug. 2022 – May 2023

Purdue University Office of Engagement

West Lafayette, IN

- Worked in a 10 person team using Agile Scrum, delivering a full stack web app for research grant applications.
- Developed RESTful API endpoints using Express.js to enable CRUD functionality and authentication.
- Implemented federated session authentication using Purdue University in-house Single Sign-On APIs.
- Architected and implemented a PostgreSQL database using Prisma ORM to handle user PII and grant applications.
- Utilized Docker for containerization and delivery of application to university server cluster.

PROJECTS

Social Network | Python, FastAPI, JavaScript, React, Docker, AWS, PostgreSQL

2024 - Present

- Collaborated with LSD.so on a decentralized social network built on the AT Protocol by Bluesky.
- Developed FastAPI/PostgreSQL backend and static React frontend on Amazon EC2 using Docker.
- Implemented ATPROTO OAuth2.0 session cookie authentication and live global WebSocket feed.
- Implemented embedded content parsing using LSD.so web SQL for user behavior data pipelines.
- Architected CI/CD pipeline with GitHub actions and AWS CodeDeploy and utilized Pytest and Jest for CI validation and integration testing.

Distributed Job Scheduler | Go, GORM, PostgreSQL, Redis, AWS, Docker, Kubernetes

2024 - 2025

- Created a distributed job scheduling platform to orchestrate jobs across nodes in a cluster environment.
- Integrated Redis for job queuing, caching, and message passing, ensuring throughput and low-latencies across a cluster.
- Developed automatic horizontal scaling mechanisms to dynamically resize worker pool via scaling action triggers and Kubernetes orchestration.
- Implemented leader election, consensus algorithms, and retry policies for fault tolerance and consistency.
- Wrote RESTful APIs for job creation, monitoring, and administrative functions via HTTP requests.